

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)			435 (3) if answer incorrect credit each correct step in method $2340 - 94 = 2246$ (1) $2246 - 941 = 1305$ (1) $H-H = 435$ (1) alternative method $2340 - 94 = 2246$ (1) $N \equiv N + 3(H-H) = 2246$ (1) $H-H = 435$ (1)			3	3	3	
	(b)			yield is lower / is too low / less ammonia is formed (1) reaction rate is lower / too low reaction is slower / is too slow (1)	2			2		
	(c)			$HNO_3 + NH_3 \rightarrow NH_4NO_3$ reactants (1) product (1)		2		2		

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	(d)			accept any of the following approaches but the second point must directly follow from the first for both marks (leads to) high numbers of algae / microorganisms (1) ⇒ decomposition of algae depletes oxygen (1) stops sunlight reaching plants below the surface (1) ⇒ plants unable to photosynthesise (1) plants unable to photosynthesise (1) ⇒ reduction in oxygen content of water (1) excessive plant growth stops absorption of oxygen (from air) ⇒ not enough oxygen for fish (1)	2			2		
				Question 7 total	4	2	3	9	3	0